Carlos Daniel Rivera Lao

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Experience

Applied Intuition Oct 2025 - Present
Software Engineer, Developer Frameworks *Sunnyvale, CA*

- Architected and developed a modular workspace platform using Bazelmod, Nix, EngFlow, and Buildkite, automating reproducible development environments, CI provisioning, and versioned tooling updates, with 10+ workspaces migrated to date as part of an ongoing organization-wide rollout.
- Designed and deployed a robust, company-wide code ownership system using Go, Python, PostgreSQL, Terraform, and AWS, establishing a centralized source of truth for 1,400+ ownership records and enabling automated ownership-based workflows.
- Built and launched a company-wide test observability platform using Python, Go, Snowflake, PostgreSQL, and Superset, developing data pipelines, dashboards, and a novel gRPC endpoint coverage metric collected via nightly CI.
- Led incident response during monthly on-call rotations and maintained infrastructure across Bazel, Buildkite, and AWS, resolving build, deployment, and developer tooling failures while implementing long-term reliability improvements.

California Costumes Aug 2024 - Oct 2025
Software Engineer *Los Angeles, CA*

- Engineered and deployed full-stack workflow automation solutions using Python and React, streamlining report generation, tracking retail metrics, and managing order routing, resulting in over 200 hours saved annually.
- Developed time-series forecasting models (Prophet) for product performance prediction, driving data-driven decision-making in sales and inventory management.
- Collaborated closely with non-technical stakeholders to gather requirements and deliver tailored software solutions, improving operational efficiency and reducing manual tasks.
- Led the planning and technical implementation of B2C infrastructure, facilitating the transition from B2B.

Meta Sep 2023 - Aug 2024
Linguistic Data Analyst *Remote*

- Designed and implemented several React-based Chrome extensions to assist red teamers in tracking productivity, monitoring task progress, and optimizing workflows.
- Stress tested various LLM models across the entire spectrum of risk categories to expose vulnerabilities and bolster system robustness and security.
- Created data pipelines for efficient annotation of over 400 data points weekly, supporting model retraining and improvements in machine learning systems.

Paywake Jun 2023 - Sep 2023
Software Engineer *Los Angeles, CA*

- Developed core internal tools for human photo verification, banning users, editing site content, and debugging.
- Programmed and refactored cross-platform, customer-facing UIs in React Native, improving performance, reducing API call overhead by 50%, and ensuring retroactive maintainability.
- Collaborated with cross-functional teams in agile sprints to troubleshoot and resolve technical issues.

Skills

Programming Languages: Python, Go, SQL, JavaScript/TypeScript, Java, C/C++, HTML/CSS, Swift

Infrastructure & DevOps: AWS, Terraform, Bazel/Bazelmod, Nix, Buildkite, EngFlow, Git, CI/CD

Frameworks & Tools: React, React Native, Next.js, Node.js, Express.js, Flask, Django, Alembic

Databases & Analytics: PostgreSQL, Snowflake, MongoDB, Firebase, Apache Superset

Spoken Languages: English, Tagalog, Mandarin, Spanish, American Sign Language

Education

University of Southern California Aug 2019 - May 2023
B.S. Computer Science, B.A. Linguistics ***GPA: 3.75/4.00***

- **Awards:** Magna Cum Laude, Dean's Scholarship, Academic Achievement Award, Dean's List
- **Thesis:** Observing Phonological and Morphological Changes of Spanish Borrowings into Filipino
- **Relevant Coursework:** Software Engineering, Algorithms and Theory of Computing, iOS App Development, Cryptography, File and Database Management, Internetworking, Natural Language Processing, Artificial Intelligence